

ERFAN ASGARI

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Research Interests

- Deep Learning
- Generative Models
- Diffusion Models
- Inpainting & Composition
- Computer Vision
- Image Generation
- Vision Language Models
- Representation Learning

Education

University of Tehran

Oct. 2020 – Present

MS. in Computer Engineering (AI and Robotics)

University of Tehran

Sep. 2020 – Sep. 2025

B.Sc. in Electrical Engineering — GPA: 18.58/20

Project: “Methods for Incorporating Reproducible Characters in Image Generating Models”

Advisor: Dr. Mostafa Tavassolipour

Research Experience

Deep Generative Models Reading Group

Dec. 2023 – Present

Active Member and Presenter, Under Supervision of Dr. Mostafa Tavassolipour Department of ECE, University of Tehran

- Reviewing papers weekly - Each member presents and leads a discussion on a chosen paper
- Presented several papers including:
 - * LoRA
 - * Flow-Matching
 - * Mamba & Vision Mamba
 - * Deep Image Composition
 - * ControlNet
 - * Self-Consuming Models
 - * Structured State Spaces
 - * ACE++

Summer Internship

Jun. 2023 – Sep. 2023

Brain-Computer Interface Developer

NBIC Lab, Tehran, Iran

- Designed and Implemented a signal acquisition board inspired by OpenBCI Ganglion, integrating updated and accessible components.

Professional Experience

Machine Learning Engineer | Medad AI

Nov. 2024 – Sep. 2025

- Deployed generative AI models for image and video, using Diffusers
- Developed and optimized computer vision pipelines, using OpenVINO and FAISS.
- Deployed STT pipelines, using NEMO and Docker.

Front-End Developer and Junior AI Engineer | Pixy

Jul 2023 – Jan 2024

- Designed and implemented a Web App for an image generation chatbot
- Fine-tuned LoRA models and developed Stable Diffusion XL pipelines with IP-Adapter and ControlNet for industrial photography.

Founder and Software Engineer | Tarhvar

Dec 2020 – Dec 2022

- Founded and led a print-on-demand merchandising platform for influencers.
- Managed a technical team, designing the database and UI/UX.

Skills

Machine Learning: PyTorch, TorchVision, TensorFlow, Diffusers, NumPy, Scikit-learn

Software Development: C, C++, JavaScript, Git, Docker, NodeJs, ReactJs, MongoDB, REST APIs

Control & Robotics: MATLAB, Simulink, PCB Design, Arduino, STM32, ROS

Hardware & Electronics: Verilog, ModelSim, Intel Quartus, Multisim, PSPICE

Additional Skills: L^AT_EX, Markdown, Image Editing, 3D Modeling, Vector Design, Animation

Honors and Awards

- Among **top 10%** of B.Sc. students in Electrical Engineering at University of Tehran.
- **Ranked 1st** in Online Preliminary Round, UTAI Hackathon (Image Processing), February 2025.
- **Ranked ~40th** in Screening Phase (among 730 teams), Rayan International AI Contest, October 2024.
- **Ranked 3rd** (among 225 teams) in *AI/Image Processing* Track, Tech-Olympics of Iran, October 2024.
- **Ranked 2nd** (among 344 teams) in *Front-end Development* Track, Tech-Olympics of Iran, October 2024.
- **Achievements in Robotics (Junior Soccer Open League ³):**
 - **1st Place for Superteam** and **3rd Place for Individual**, Robocup Asia-Pacific 2018 - Kish, Iran
 - **1st Place for Superteam**, Robocup Asia-Pacific 2017 - Bangkok, Thailand
 - **1st Place for Superteam** in 4 national competitions

³In the Junior Soccer Open-Weight League, robots use vision systems to detect the ball and goals.

Highlighted Grades

- **Linear Algebra:** 19/20
- **Intelligent Systems:** 20/20
- **Machine Learning (Graduate):** 20/20
- **Deep Generative Models⁴ (Graduate):** 20/20
- **Advanced Programming** 19.9/20
- **Operations Research:** 19.8/20

Teaching Assistantship

- **Machine Learning** Spring 2024 – Spring 2025
Assignment Designer and Grader
- **Advanced Programming** Fall 2023 – Spring 2025
Website Developer, Assignment Designer and Grader
- **Linear Algebra** Fall 2023 – Spring 2024
Computer Assignment Designer and Grader
- **Deep Generative Models** Fall 2024
Assignment Designer and Grader

Selected Course Projects

Deep Generative Models | *PyTorch, TorchVision, Diffusers, Transformers*

- Developed a **VAE** using Negative ELBO and Negative IWAE to learn image latent representations.
- Trained a **Normalizing Flow** model based on **RealNVP** to generate images.
- Developed a **GAN** model, evaluated using **FID Score**, and enhanced with **Wasserstein distance**.
- Trained a **Diffusion model** for image generation based on **DDPM** paper on the MNIST dataset.
- Evaluated an **LLM** for zero-shot, few-shot, and chain-of-thought reasoning, and improved by **LoRA** training.
- Fine-tuned SpeechT5 for **Speech Synthesis** at the character level by adding new characters and training on the CommonVoice dataset to enable reading Persian texts.

Machine Learning & Intelligent Systems | *NumPy, Scikit-Learn, TensorFlow*

- Developed **Fake Image Detection** using multiple **feature-extraction**, **supervised** and **unsupervised** techniques.
- Implemented **Deep Squish Predictor** to recover original aspect ratio of images using **self-supervised** techniques.
- **Reinforcement Learning** (policy and value iteration), and **Deep Q-Learning** on Gymnasium Breakout.
- Collected an image dataset and trained a **CNN** model from scratch for robot type **classification**.
- Fine-tuned **YOLOv8** for object detection and **SAM** for segmentation on custom dataset.

Linear Algebra | *MATLAB, Python*

- Developed an **SVD-based Image Watermarking** technique for multimedia protection.
- Analyzed business cycles using **Markov Chains** and **Eigenvalue Decomposition**.
- Applied **LSA** (Latent Semantic Analysis) for news classification using **Truncated SVD** and **Similarity Metrics**.

Mechatronics | *MATLAB, Simscape, ROS, Python*

- Estimated rotation matrix using **IMU Sensor** and **Kalman Filter** and pose detection on video using **MediaPipe**.
- Analyzed the kinematics and dynamics of a **SCARA** robot using MATLAB and **Simscape** with a Simulink model.
- Developed a **Publisher-Subscriber Node** and launch application for controlling TurtleSim in **ROS**.

Control Engineering | *MATLAB, Simulink*

- Developed and analyzed a **Magnetic Levitation System**, incorporating system linearization, stability analysis, and designing PID and PI controllers
- Designed a **Ball and Beam Control System**, with a focus on stabilization, employing PID controllers, state feedback, and observers.

Online Courses

- **Deep Learning Specialization:** 5 Courses [[1](#), [2](#), [3](#), [4](#), [5](#)]
 - **Mathematics for Machine Learning:** [[Linear Algebra](#)]
 - **Zero to Mastery PyTorch for Deep Learning**
 - **Hugging Face Diffusion Models Course**
- DeepLearning.AI
Imperial College London
LearnPyTorch.io
Hugging Face

Languages

- **English:** Professional Working Proficiency
IELTS: 7.5 - Listening (8), Reading (8), Writing (6.5), Speaking (7)
- **Persian:** Native

⁴Listed as 'Advanced Topics in Control 1' on the transcript